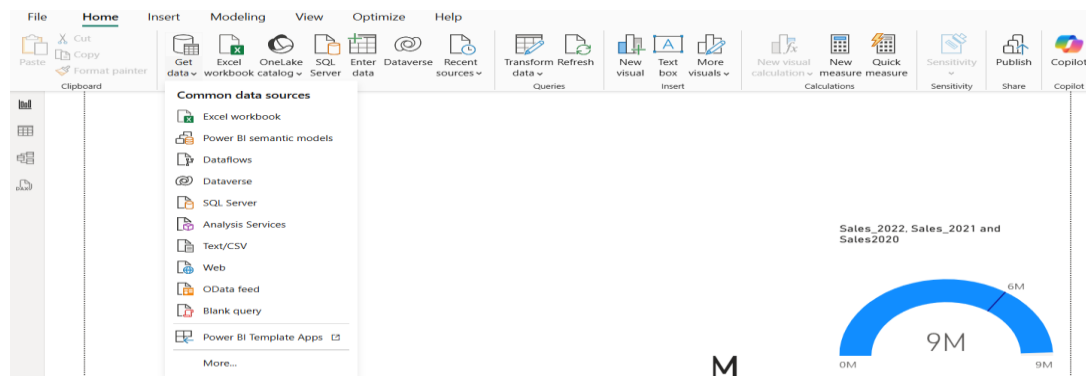


DATA MODELLING IN POWER BI

What is **Data Modeling** in Power BI?

Data modeling is the process of:

- Importing data from various sources (Excel, SQL Server, SharePoint, etc.)
- Transforming and cleaning that data (using Power Query)
- Creating relationships between different tables
- Defining measures, calculated columns, and hierarchies using DAX (Data Analysis Expressions)
- Designing a structure that makes reporting easy, accurate, and efficient



◆ Components of Data Modeling in Power BI

1. Tables

- Raw data from sources
- Can be fact tables (e.g., Sales) or dimension tables (e.g., Date, Product)

2. Relationships

- Links between tables
- Usually defined using primary key and foreign key
- Can be:
 - One-to-many (most common)
 - Many-to-many
 - One-to-one
- Helps Power BI understand how data connects (e.g., which customer bought which product)

3. Star Schema vs Snowflake Schema

- Star schema is preferred in Power BI (Fact table at center, surrounded by dimension tables)
- It improves performance and usability

4. Calculated Columns

- New columns created using DAX
- Stored in the model (e.g., Profit = Sales - Cost)

5. Measures

- Calculations that are computed on the fly (dynamic)
- Very efficient and powerful
- Example: Total Sales = SUM(Sales[Amount])

6. Hierarchies

- Logical grouping, like Year > Quarter > Month > Day
- Used for easy navigation and drill-down in visuals

7. Data Types and Formatting

- Ensuring fields are correctly typed (e.g., date, number, text)
- Important for proper visuals and calculations

